

Split

$s = \text{"hi my name is Rahim"}$

`s.split()`

`['hi', 'my', 'name', 'is', 'Rahim']`

Join

`"".join(['hi', 'my', 'name', 'is', 'Rahim'])`

Two pointers approach

$s = \text{C i v i c}$

$s[i] == s[j]$

$i < j$ $i > j$

break

$s[i] != s[j]$

Not palindrom

break

$j = \text{len}(s) - i - 1$

C i v i c
0 1 2 3 4

C i v i c

$s[i] != s[j]$

False

False

False

flag

True

T

T

for i in range($\text{len}(s) // 2$)

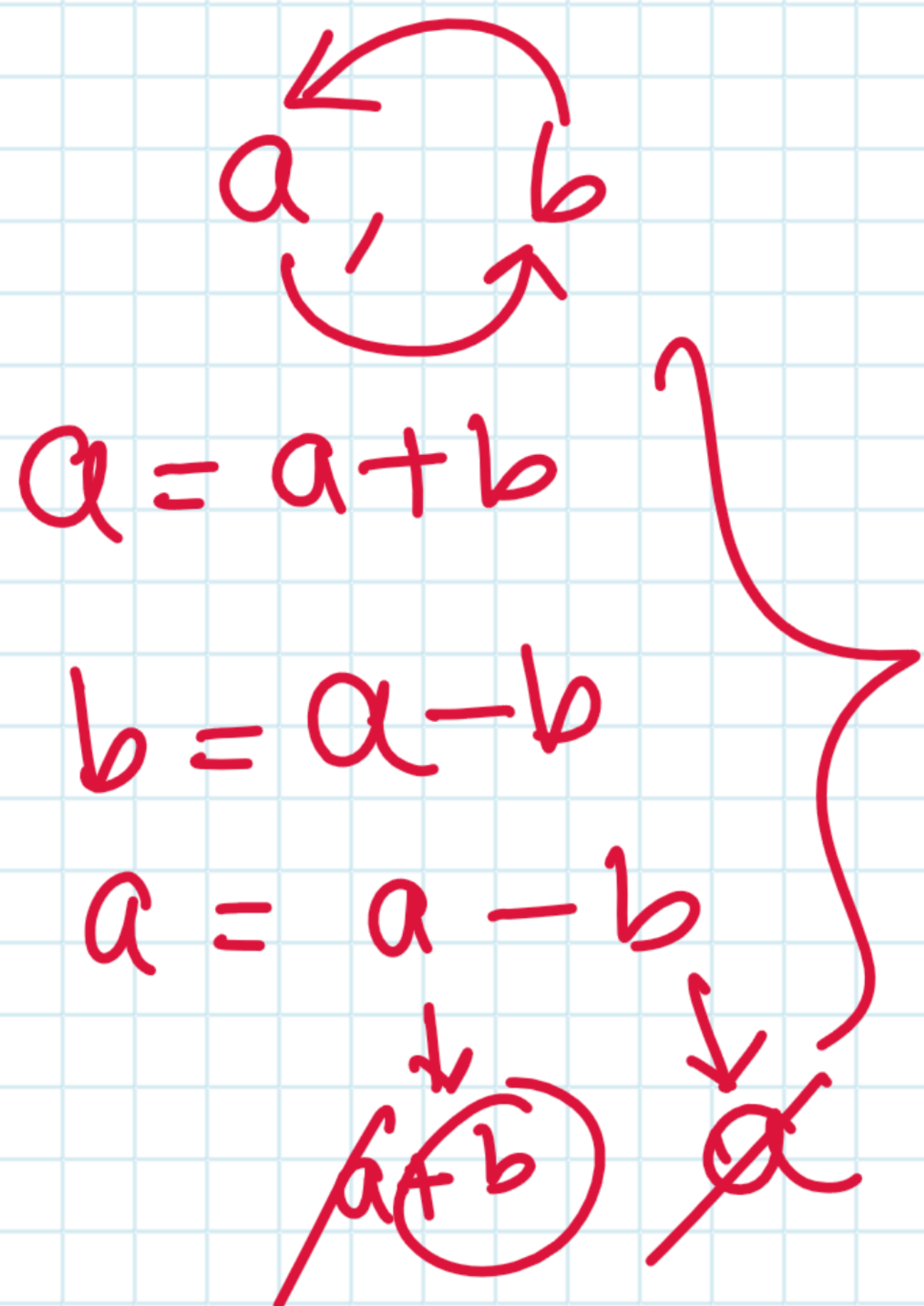
Task

Take two numbers as input and swap them

$a = 4$
 $b = 5$

$a = 5$
 $b = 4$

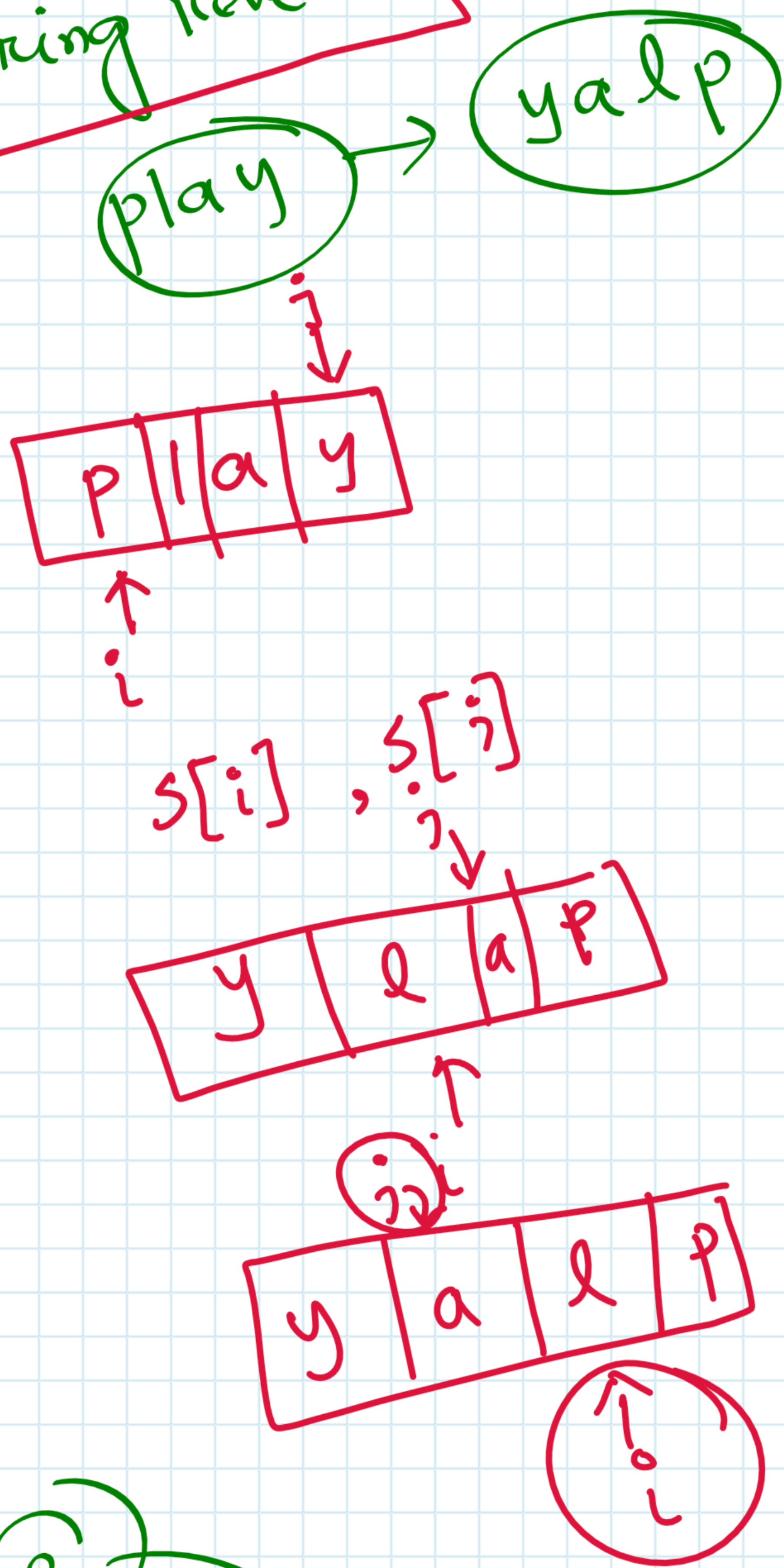
$a, b = b, a$ python syntax



$first = 0$
 $second = 1$
 $second = second + first$
 $first = second$

"76432"

String Reverse



$76432 \% 10 = 2$
 $76432 // 10 = 7643$
 $7643 \% 10 = 3$
 P_i

1000
 $temp = 1000$
 $digit = 0$
 $temp = 100$
 $digit = 0$
 $temp = 10$
 $digit = 0$
 $temp = 1$

```

L=[]
for number in range(1000,3001):
    flag=True
    temp=number
    while temp!=0:
        digit=number%10
        temp//=10
        if digit%2==1:
            flag=False
            break
    if flag:
        L.append(number)
print(L)

```

"Prime numbers"

number

1

number

Hello

1234

Hello

1 to 35

35
1x35
5x7

8
1x8
2x4

16
1x16
2x8
4x4

"Number theory"

Prime

2 to

number-1

if divisible
Not prime

7
2 → 6
3, 4, 5, 6

Lists

L = [20, 'short', 35.75, [30, 40, 50]]
↑ ↑ ↑ ↑
L[0] L[1] L[2] L[3]

Why Lists?

Array vs List

fixed

Dynamic

int a[5]

a → 0 1 2 3 4

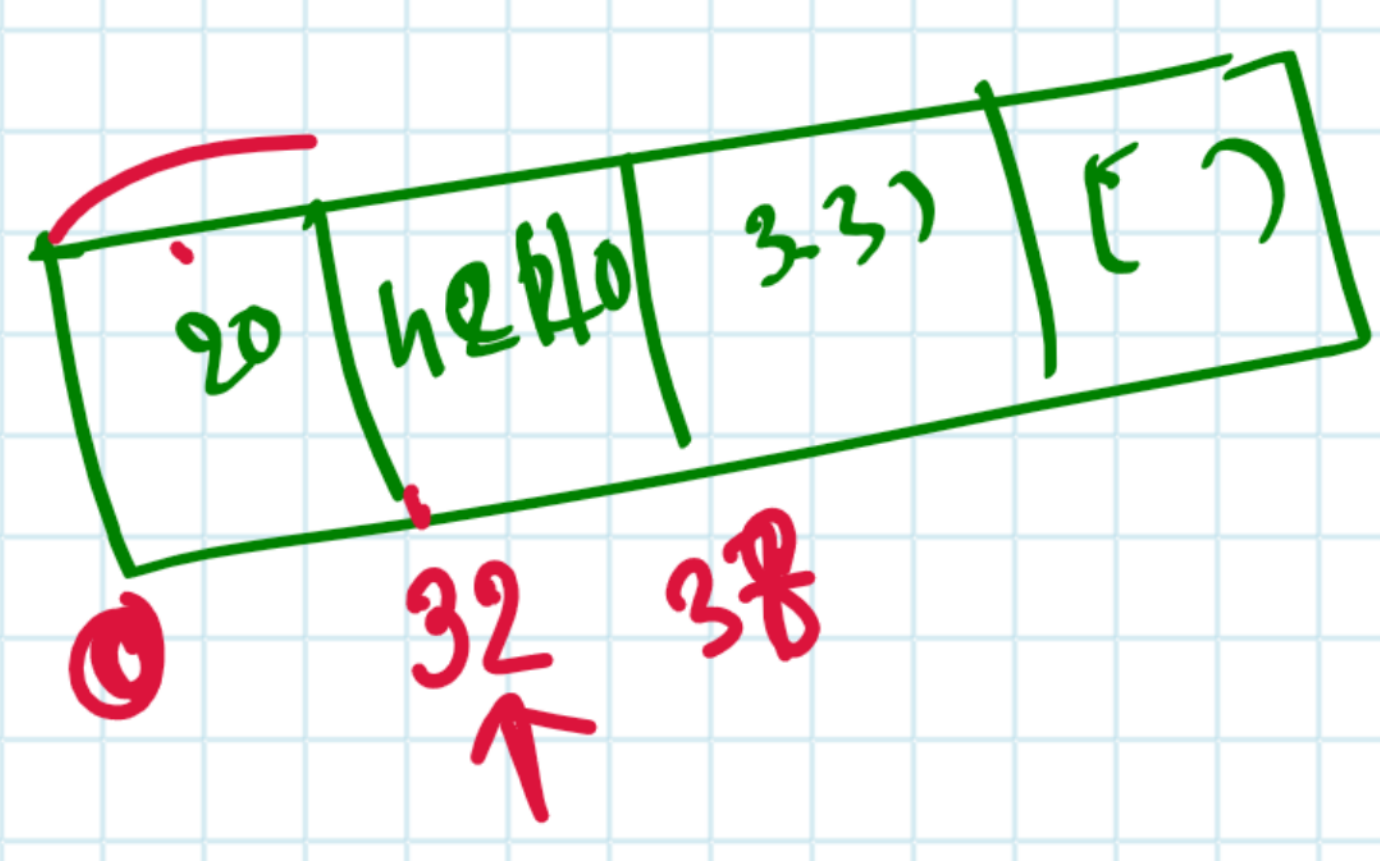
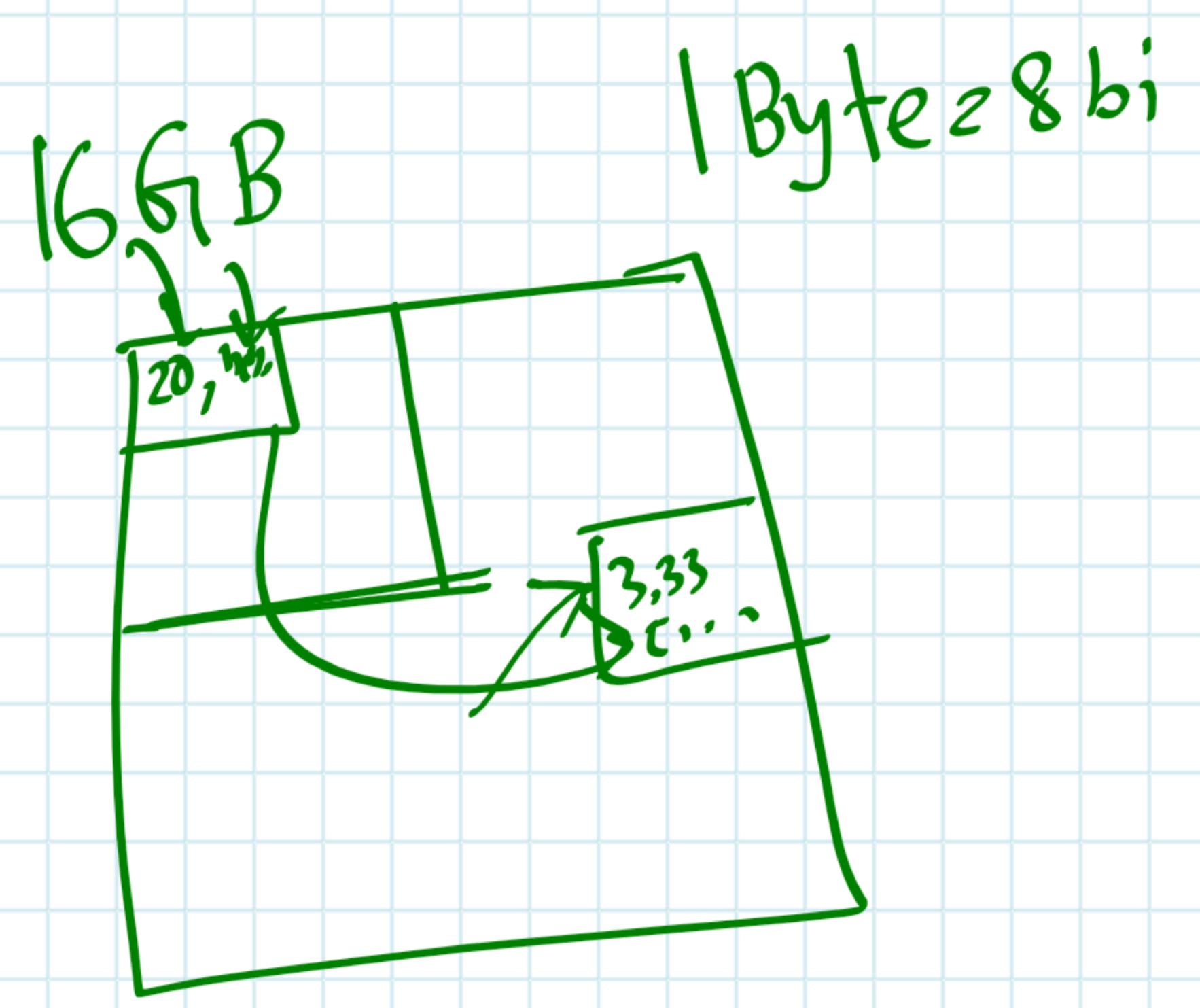
Array vs List

convenience: Heterogeneous

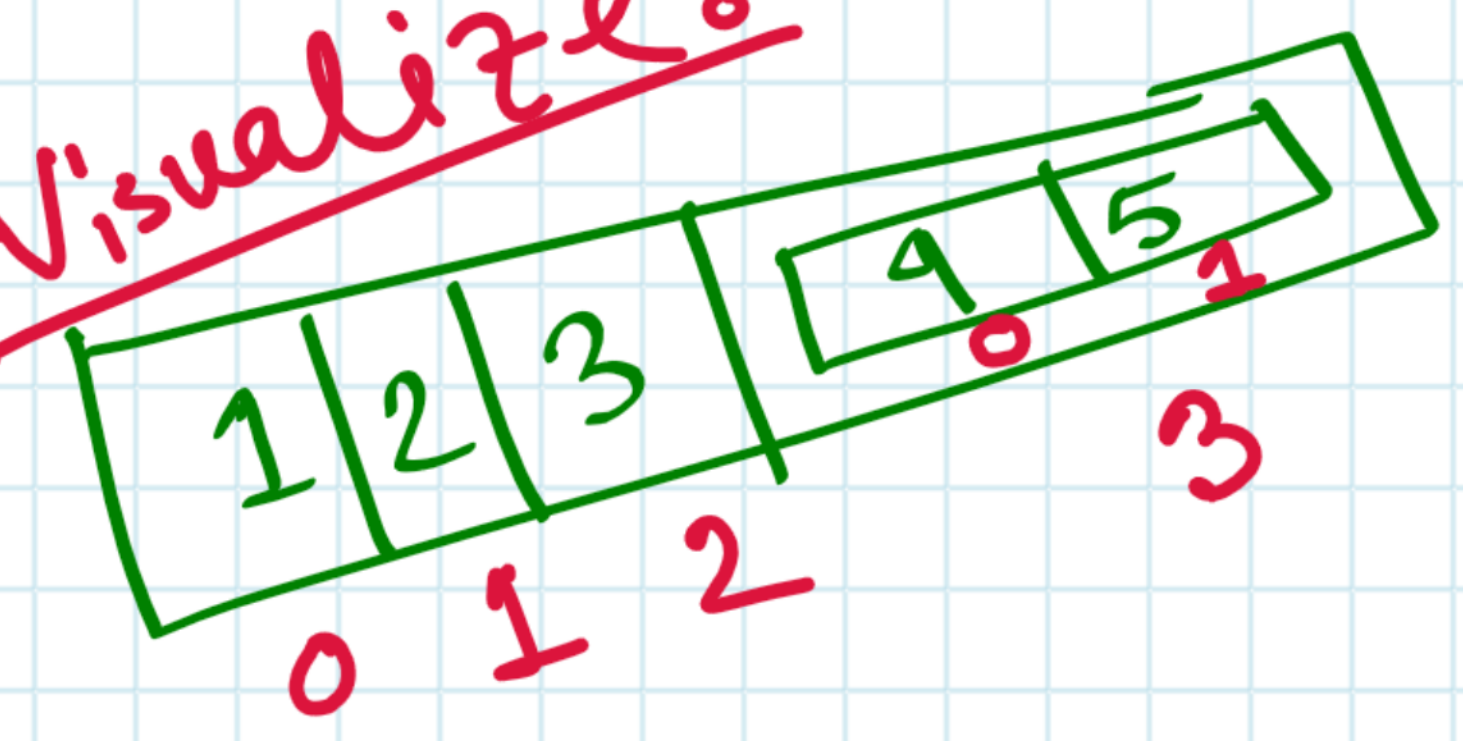
int a[5]
string a[5]
char a[5]

L = [20, 'hello', 3.33, [1,2,3]]

RAM
ROM
HDD



Visualize:



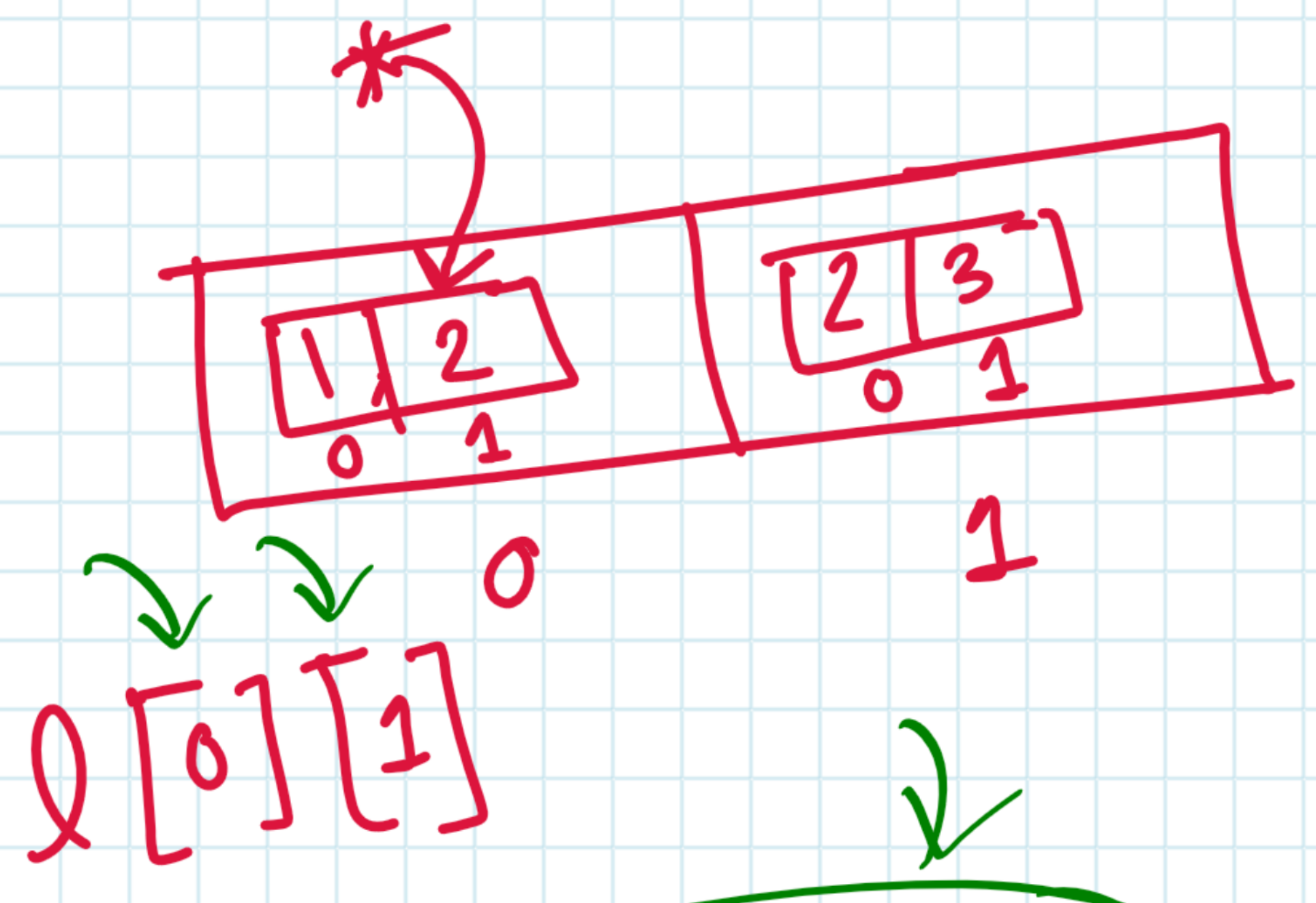
List_2D_demo = [1, 2, 3, [4, 5]]

List_2D_demo[3] ~> [4, 5]

List_2D_demo[3][0] ~> 4

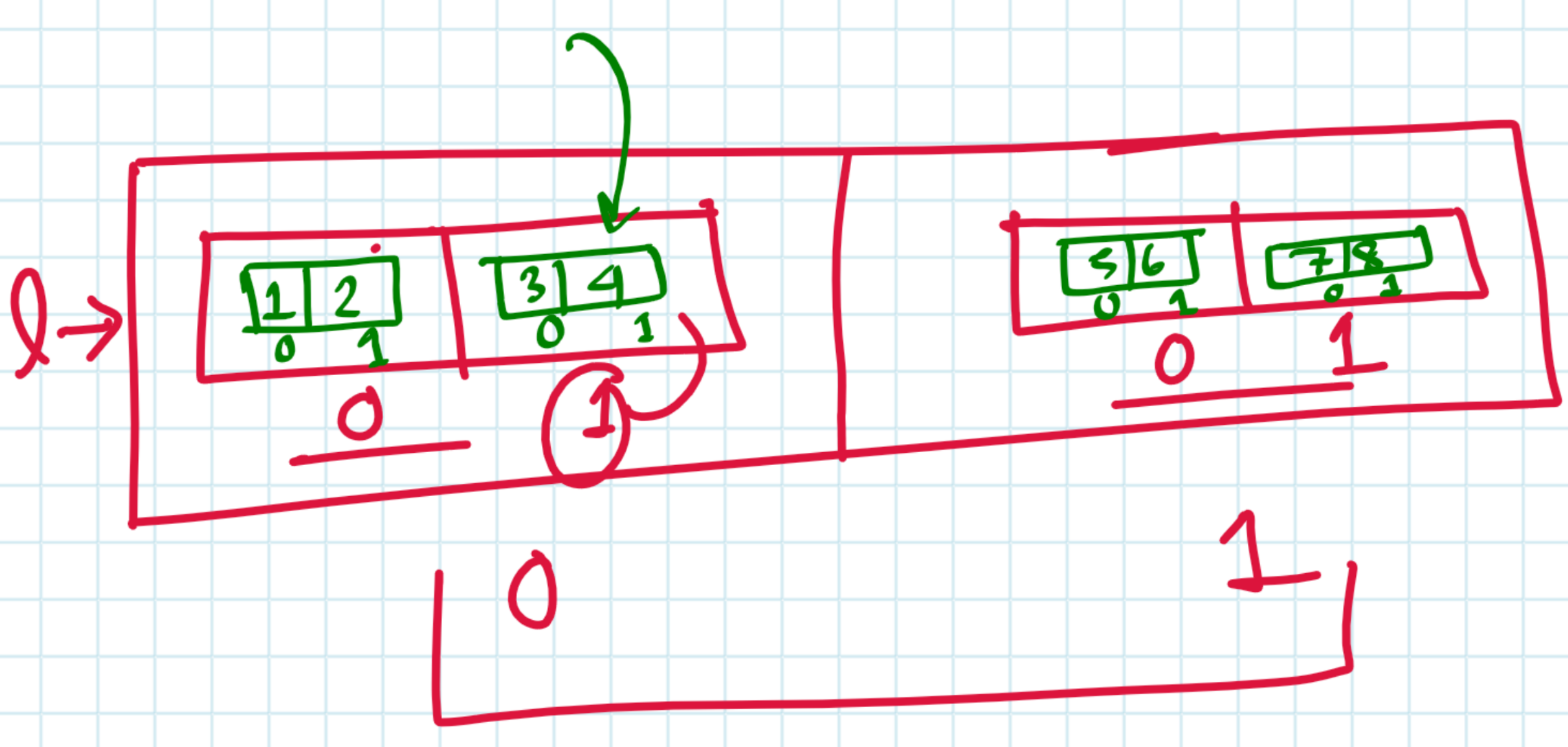
List_2D_demo[3][1] ~> 5

L -> [[1,2], [2,3]]



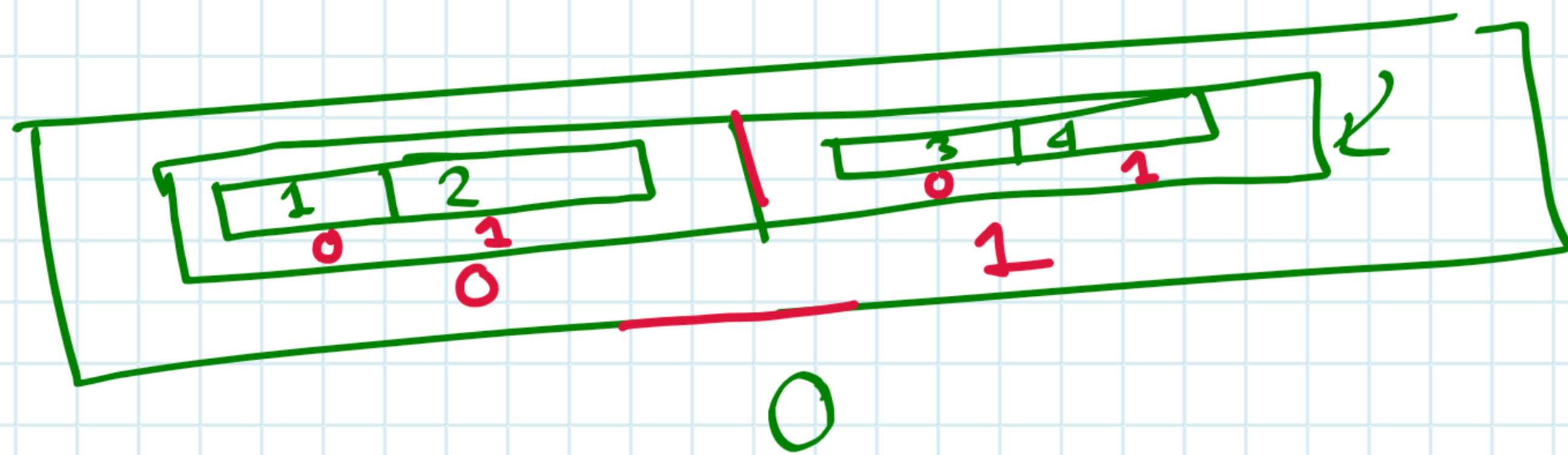
L[0][1]

3D L[0][1][0]



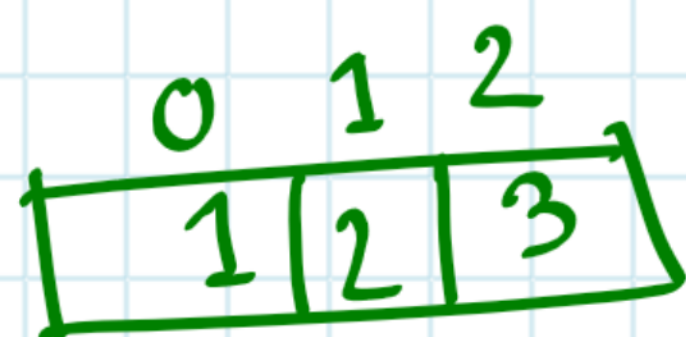
L[0][1][1] -> 4

$[[[1,2], [3,4]]]$

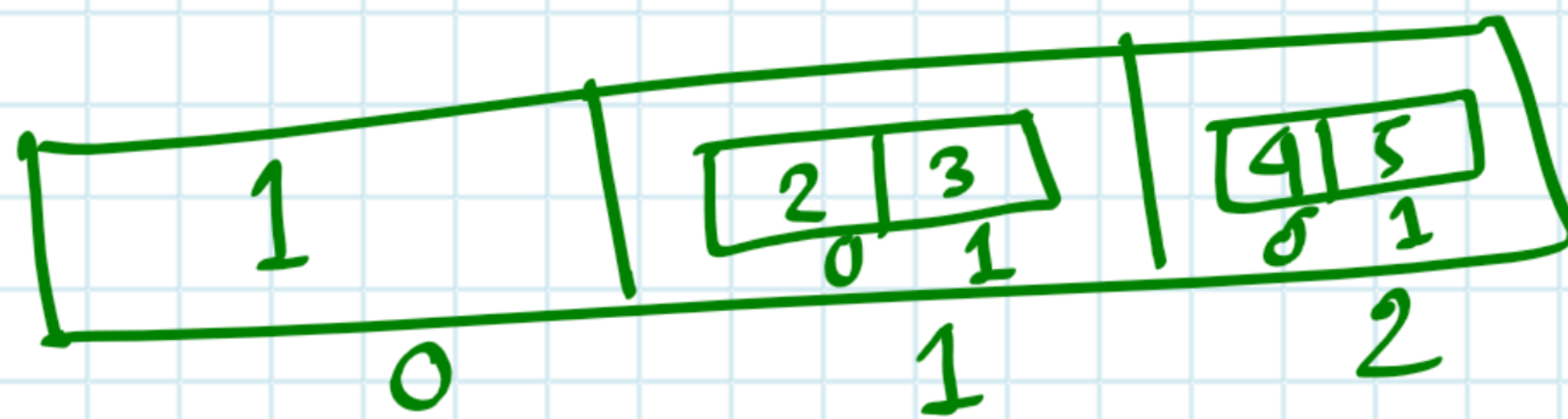


$[[[1,2], [3,4]], [[5,6], [7,8]]]$

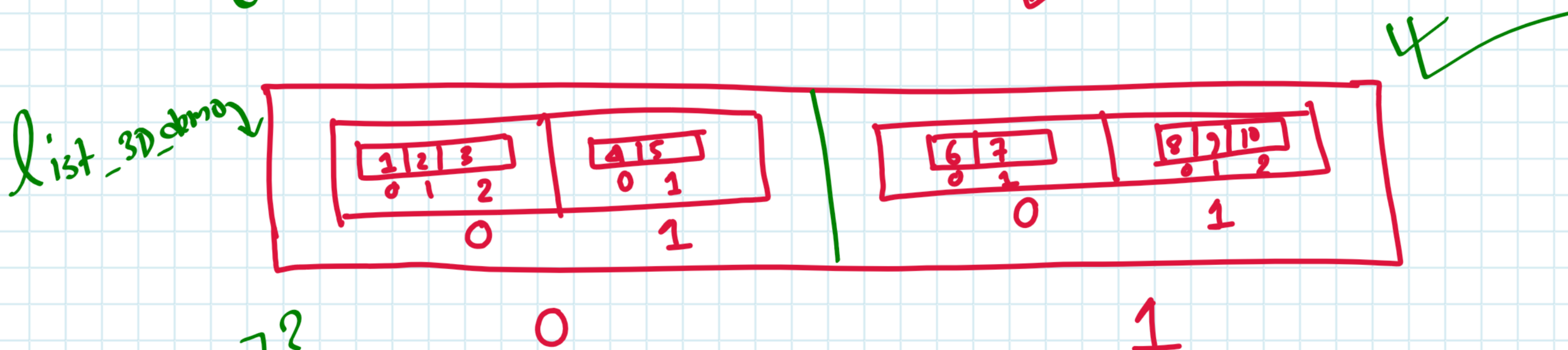
$[1,2,3] \rightarrow \checkmark$



$[1, [2,3], [4,5]] \rightarrow \checkmark$



$[[[1,2,3], [4,5]], [[6,7], [8,9,10]]]$



How to access 7?

$list_3D_demo[1][0][1] \rightarrow 7$

